
COURSE MANAGEMENT SYSTEM

COURSE NAME

SOFTWARE ENGINEERING (CS-321)

SEMESTER

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1. Introduction

The **Online Course Registration System** is a web-based academic management platform designed to simplify and automate the process of course enrollment, subject management, and fee processing within educational institutions. The system provides an efficient environment where students can register for courses online, professors can manage course-related information, and administrators can monitor academic operations effectively.

Traditional course registration methods in many institutions are still partially manual or dependent on outdated systems that require significant administrative effort. These methods often create delays, data inconsistencies, scheduling conflicts, and inconvenience for both students and faculty members. To overcome these issues, the proposed Online Course Registration System offers a centralized digital solution that improves accessibility, accuracy, and operational efficiency.

The system is mainly intended for:

- **Students**, who need to browse available subjects, register courses, view schedules, and complete fee payments.
- **Professors**, who need to manage courses, upload materials, and monitor enrolled students.
- **University Administration**, responsible for maintaining academic records, registration policies, and billing coordination.

The proposed system aims to provide a secure, user-friendly, and reliable platform that supports real-time data management and smooth communication among all stakeholders.

What

The Online Course Registration System is a software application developed to automate academic registration activities within a university or educational institute. The system allows students to select and register courses online without requiring physical paperwork or manual approvals in most cases.

The software includes several core functionalities such as:

- Student authentication and login
- Course browsing and selection
- Automatic prerequisite checking

- Credit hour validation
- Online fee payment integration
- Course management by professors
- Real-time seat availability updates
- Academic record management

The system is designed to reduce manual workload, minimize registration errors, and improve the overall academic registration experience.

Who

The primary users and stakeholders of the system include:

1. Students

Students are the main users of the system. They can:

- Log into their accounts
- View available courses
- Register or drop subjects
- Check prerequisites
- Make online payments
- View registration history and schedules

2. Professors

Professors use the system to:

- Manage assigned courses
- Upload course materials
- View enrolled student lists
- Update course-related information

3. University Administration

Administrative staff are responsible for:

- Managing student records
- Monitoring registrations

- Controlling course offerings
- Supervising payment verification
- Maintaining academic policies

4. Billing Department

The billing system or finance department processes:

- Student fee payments
- Payment verification
- Billing records
- Financial reporting

Problems

The existing manual or semi-automated registration systems face several operational and technical challenges. These problems affect both students and university management.

The major problems identified are:

1. Manual Registration Delays

Traditional registration systems require physical form submission or lengthy approval procedures, which consume significant time and create delays during enrollment periods.

2. Data Redundancy and Errors

Manual data handling increases the chances of duplicate records, incorrect course assignments, and human errors in student information management.

3. Lack of Real-Time Updates

Students may not receive accurate information regarding:

- Available seats
- Course schedules
- Registration deadlines
- Payment confirmations

This often creates confusion and registration conflicts.

4. Difficulty in Managing Prerequisites

Existing systems may not properly verify course prerequisites, resulting in students enrolling in subjects for which they are not eligible.

5. Inefficient Communication

Coordination between students, professors, and administration becomes difficult when information is spread across multiple platforms or handled manually.

6. Security Concerns

Older systems may lack secure authentication and payment mechanisms, increasing risks related to unauthorized access and data breaches.

7. Heavy Administrative Workload

University staff often spend excessive time handling registration forms, updating records, and resolving scheduling conflicts manually.

1.1 Existing Systems

Several universities currently use traditional or partially computerized registration systems. These systems generally provide basic registration functionality but still suffer from limitations in automation, usability, and integration.

Common existing systems include:

- Manual paper-based registration systems
- Spreadsheet-based student management
- Basic web portals with limited functionality
- Standalone departmental registration software

In many institutions, students must physically visit departments to complete registration procedures, submit documents, or verify payments. Some institutions provide online registration portals; however, these systems may not support advanced features such as:

- Automatic prerequisite validation
- Real-time seat tracking
- Integrated online payments
- Dynamic scheduling
- Centralized academic management

Most existing systems also have limited scalability and become inefficient when handling large numbers of users simultaneously during semester registration periods.

1.2 Comparative Analysis

The proposed Online Course Registration System improves existing systems by providing a centralized, automated, and secure platform for academic registration management.

Feature	Existing Systems	Proposed System
Registration Process	Mostly manual or semi-automated	Fully online and automated
Course Availability Updates	Delayed or manual	Real-time updates
Prerequisite Checking	Manual verification	Automatic validation
Payment Processing	Separate or offline	Integrated online billing
User Accessibility	Limited access hours	24/7 online access
Data Accuracy	Higher chances of errors	Improved accuracy through automation
Security	Basic authentication	Secure login and payment handling
Administrative Workload	High	Reduced through automation
Scalability	Limited	Supports large numbers of users
User Experience	Complex and time-consuming	User-friendly and efficient

The proposed system offers better performance, reliability, and usability compared to traditional registration methods. It minimizes human effort, improves data integrity, and enhances communication between students, professors, and administrators.

2. System Modules

The Online Course Registration System is divided into several functional modules to ensure proper organization, maintainability, and efficient management of academic activities. Each module is responsible for handling a specific set of operations within the system. These modules work together to provide a smooth and integrated experience for students, professors, and administrators.

The modular structure of the system improves scalability, simplifies future enhancements, and reduces system complexity.

2.1 Modules

1. User Authentication Module

The User Authentication Module manages user login, logout, and account verification processes. It ensures that only authorized users can access the system according to their assigned roles.

Main Functions:

- User login and logout
- Password verification
- User session management
- Role-based access control
- Secure authentication

Users:

- Students
- Professors
- Administrators

2. Student Management Module

This module manages student-related information and academic records. It stores and updates student profiles and registration details.

Main Functions:

- Student profile management
- Academic information storage
- Course history management
- Registration tracking

- Credit hour monitoring

Users:

- Students
- Administration

3. Course Registration Module

The Course Registration Module is the core component of the system. It allows students to register, add, or drop courses online.

Main Functions:

- Display available courses
- Course selection and registration
- Add/drop functionality
- Seat availability checking
- Registration confirmation

Users:

- Students

4. Prerequisite Validation Module

This module automatically checks whether students meet course eligibility requirements before registration.

Main Functions:

- Prerequisite checking
- Eligibility verification
- Academic rule enforcement
- Credit hour validation

Users:

- Students
- Administration

5. Professor Management Module

The Professor Management Module allows faculty members to manage course-related activities and student information.

Main Functions:

- View enrolled students
- Upload course materials
- Update course details
- Manage announcements

Users:

- Professors

6. Billing and Payment Module

This module handles fee calculation, payment processing, and billing integration.

Main Functions:

- Fee generation
- Online payment processing
- Payment verification
- Billing record maintenance
- Transaction history management

Users:

- Students
- Billing Department
- Administration

7. Notification Module

The Notification Module informs users about important academic updates and system activities.

Main Functions:

- Registration alerts
- Payment confirmations
- Schedule updates

- Deadline reminders
- System notifications

Users:

- Students
- Professors
- Administration

8. Database Management Module

This module manages all system data storage and retrieval operations.

Main Functions:

- Store user information
- Maintain course records
- Preserve transaction history
- Ensure data integrity
- Backup and recovery management

Users:

- System Administration

2.2 Stakeholders and Actors

Stakeholders and actors are individuals or groups who interact with the system directly or indirectly. They play important roles in ensuring successful operation and management of the Online Course Registration System.

Stakeholders

1. Students

Students are the primary stakeholders because the system is mainly developed to facilitate their academic registration process.

2. Professors

Professors use the system to manage course-related activities and student records.

3. University Administration

The administration supervises academic operations, manages records, and enforces institutional policies.

4. Billing Department

The billing department is responsible for handling fee processing and payment verification.

5. System Developers

Developers are responsible for system design, implementation, maintenance, and future upgrades.

6. University Management

University management monitors the overall performance and effectiveness of the system.

Actors

Actors are entities that directly interact with the system.

Actor	Description
Student	Registers courses, views schedules, and makes payments
Professor	Uploads course details and manages course information
Administrator	Manages users, courses, and academic records

2.3 Roles and Responsibilities

Each stakeholder and actor has specific responsibilities within the system to ensure smooth operation and effective academic management.

1. Student

Roles:

- Register courses
- Drop courses
- View available subjects
- Check schedules
- Make fee payments

Responsibilities:

- Provide accurate registration information
- Follow registration deadlines
- Meet prerequisite requirements
- Maintain account security

2. Professor

Roles:

- Manage assigned courses
- Upload course materials
- View enrolled students
- Update course information

Responsibilities:

- Maintain updated course details
- Upload academic resources on time
- Monitor student enrollment

3. Administrator

Roles:

- Manage users and records
- Control course offerings
- Supervise registration activities
- Generate reports

Responsibilities:

- Maintain system accuracy
- Ensure policy enforcement
- Resolve registration conflicts
- Monitor system performance

4. Billing Department

Roles:

- Verify payments
- Process billing records
- Manage financial transactions

Responsibilities:

- Ensure secure payment processing
- Maintain transaction records
- Handle billing-related issues

5. System Developers**Roles:**

- Design and develop the software
- Maintain system functionality
- Implement security measures

Responsibilities:

- Fix system bugs
- Improve system performance
- Ensure system reliability
- Provide technical support

6. Database System**Roles:**

- Store academic and user data
- Maintain data consistency
- Support system operations

Responsibilities:

- Ensure data integrity
- Prevent data loss
- Perform backup and recovery operations

3. Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) document defines the complete functional and non-functional requirements of the Online Course Registration System. It serves as a formal agreement between stakeholders, developers, and users regarding the expected behavior and capabilities of the software system.

The purpose of this SRS is to:

- Clearly define system requirements
- Identify user interactions with the system
- Describe system functionalities
- Provide a foundation for system design and development
- Reduce misunderstandings between stakeholders and developers

The Online Course Registration System is designed to automate academic registration activities while improving efficiency, accuracy, and accessibility for students, professors, and administrators.

3.1 Questionnaire / Data Collection

To gather accurate requirements for the Online Course Registration System, different requirement collection techniques were used. Data was collected from students, professors, and administrative staff to understand the existing registration process and identify its weaknesses.

The main purpose of the questionnaire and data collection process was:

- To identify user needs and expectations
- To analyze problems in existing systems
- To understand academic workflow
- To gather suggestions for system improvement

Data Collection Techniques

1. Questionnaires

Questionnaires were distributed among students and faculty members to gather feedback regarding:

- Ease of registration
- System usability
- Common registration issues
- Desired features

2. Document Analysis

Existing registration forms, academic policies, and university regulations were studied to understand:

- Credit hour policies
- Course prerequisite structures
- Registration deadlines
- Billing procedures

Sample Questionnaire

Questions for Students

1. Do you face difficulties during course registration?
2. Is the current registration process time-consuming?
3. Would you prefer online fee payment integration?
4. Do you receive real-time updates about seat availability?
5. Have you experienced prerequisite-related issues before?

Questions for Professors

1. Is it easy to manage student enrollments in the current system?
2. Do you need an online platform for uploading course materials?
3. Are registration records managed efficiently?
4. Would automated enrollment reports improve workflow?

Questions for Administration

1. Does the current system reduce administrative workload?
2. Are registration conflicts handled efficiently?
3. Is data management secure and reliable?

4. Are payment records properly integrated with registration data?

3.2 Functional Requirements

Functional requirements describe the specific operations and services that the Online Course Registration System must perform.

1. User Authentication

- The system shall allow users to log in using valid credentials.
- The system shall support role-based access control.
- The system shall allow users to log out securely.

2. Student Registration

- The system shall allow students to view available courses.
- The system shall allow students to register for courses online.
- The system shall allow students to add or drop courses.
- The system shall display course schedules and seat availability.

3. Prerequisite Verification

- The system shall automatically validate course prerequisites.
- The system shall restrict registration for ineligible students.

4. Credit Hour Validation

- The system shall enforce minimum and maximum credit hour limits.
- The system shall notify students if limits are exceeded.

5. Payment Processing

- The system shall generate fee invoices automatically.
- The system shall support online payment processing.
- The system shall verify payment completion before final registration.

6. Professor Management

- The system shall allow professors to upload course materials.
- The system shall allow professors to view enrolled students.
- The system shall allow professors to update course details.

7. Notification System

- The system shall send notifications regarding:
 - Registration confirmation
 - Payment status
 - Schedule updates
 - Registration deadlines

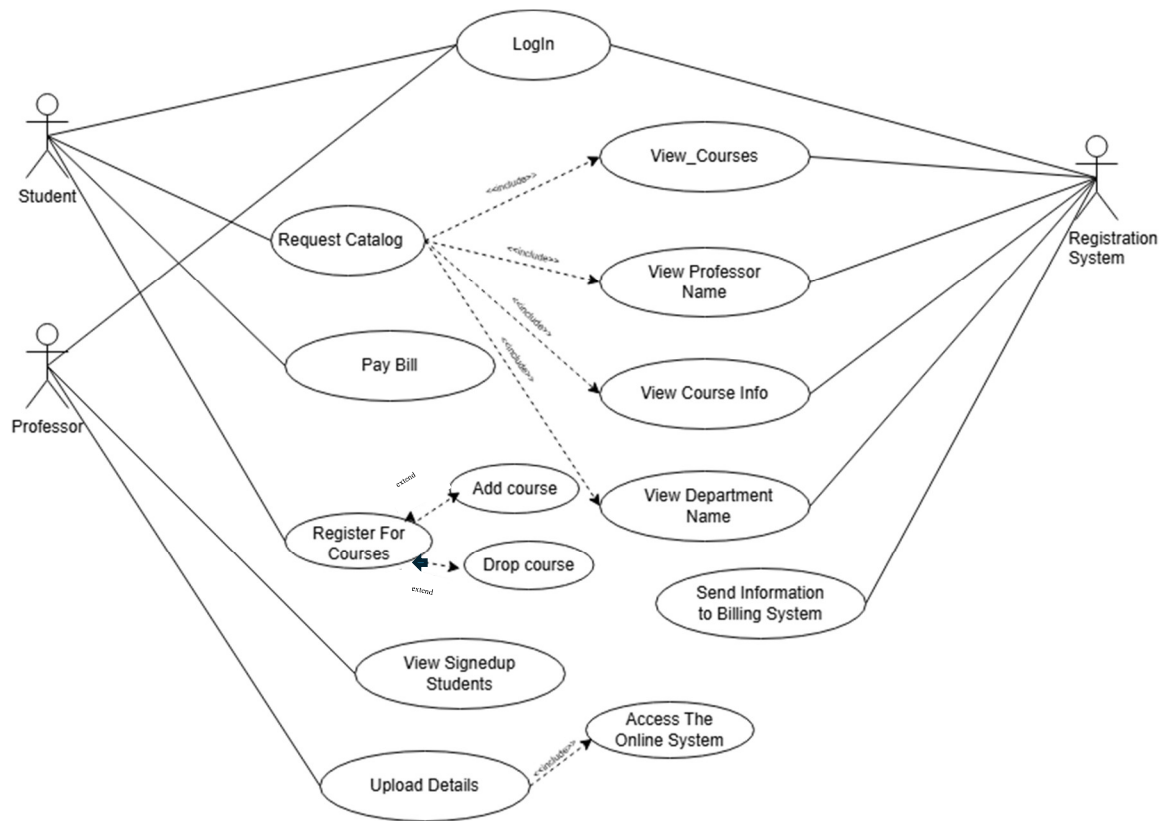
8. Data Management

- The system shall store student and course records securely.
- The system shall maintain data consistency and integrity.
- The system shall provide backup and recovery support.

9. Reporting

- The system shall generate registration reports.
- The system shall generate payment summaries.
- The system shall allow administrators to monitor enrollment statistics.

3.3 Use Case Diagram



3.4 Use Case Description

Use Case 1: Register for Courses

Name

Register Course

Summary

This use case allows students to register for available courses online.

Actor

Student

Dependency

Login

Trigger Condition

The student accesses the registration dashboard.

Pre-Condition

- Student must be logged into the system.
- Registration period must be active.

Main Flow (Normal Scenario)

1. Student logs into the system.
2. Student opens the course registration section.
3. System displays available courses.
4. Student selects desired courses.
5. System validates prerequisites and seat availability.
6. Student confirms registration.
7. System updates enrollment records.
8. Registration confirmation is displayed.

Alternative Flow

- If the course is full, the system offers a waitlist option.
- If prerequisites are not satisfied, the system suggests alternative courses.

Exceptional Flow

- If the system crashes, the registration process is interrupted.
- If the user session expires, the student must log in again.

Post-Condition

- Selected courses are successfully registered.
- Student enrollment record is updated.

Use Case 2: Upload Course Details

Name

Upload Course Details

Summary

This use case allows professors to upload and manage course-related information.

Actor

Professor

Dependency

Login

Trigger Condition

Professor accesses the course management section.

Pre-Condition

- Professor must be logged into the system.
- Professor must be assigned to the selected course.

Main Flow (Normal Scenario)

1. Professor logs into the system.
2. Professor selects a course.
3. Professor uploads course materials or updates course information.
4. System validates the uploaded content.
5. System saves the uploaded details successfully.

Alternative Flow

- If upload speed is slow, the system allows retry attempts.
- Existing course details may be edited or replaced.

Exceptional Flow

- If file format is unsupported, an error message is shown.
- If the internet connection fails, upload is paused until reconnection.

Post-Condition

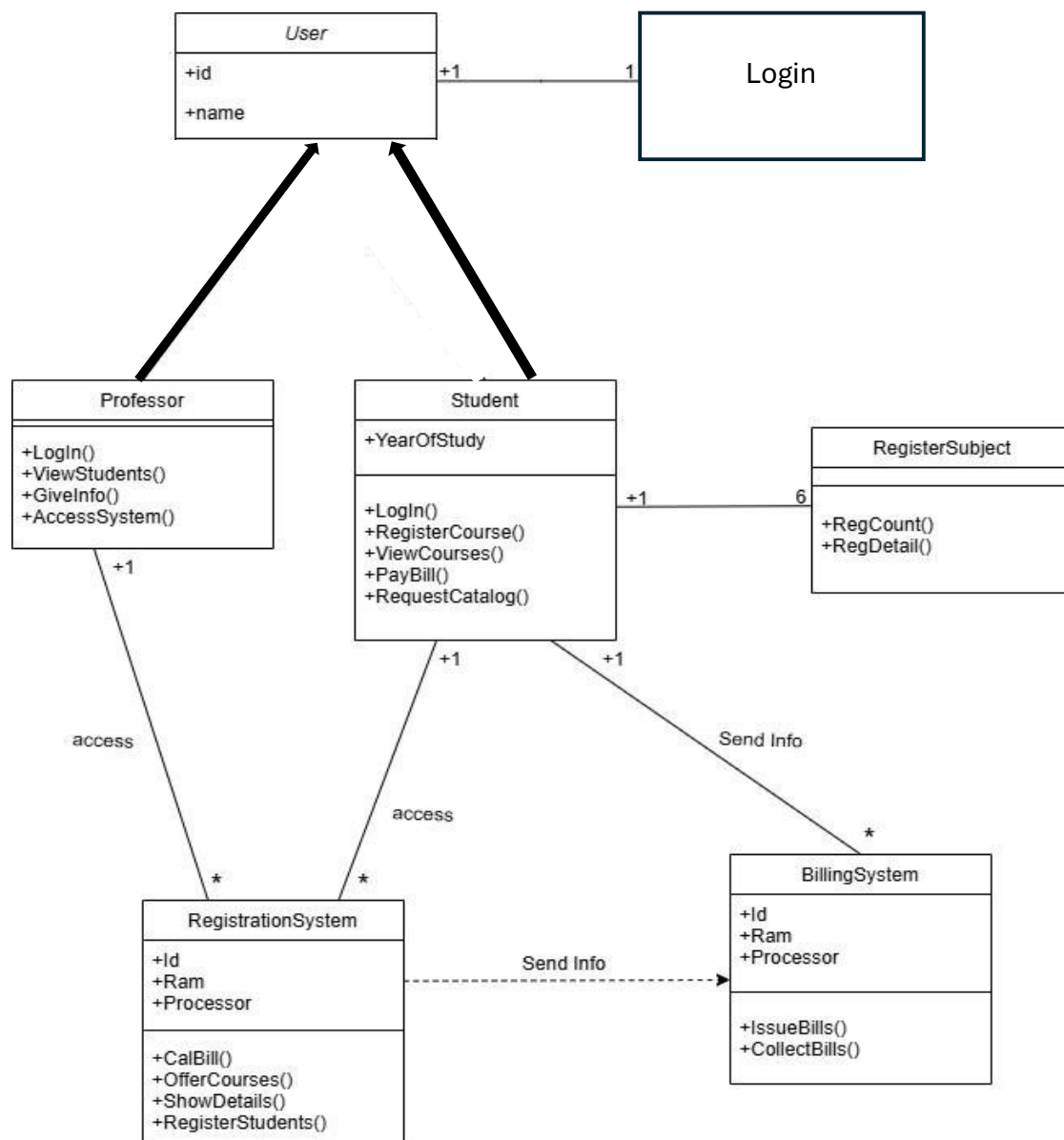
- Course details are successfully uploaded and stored in the system database.

4. Software Design

Software design defines the structure, architecture, and interaction of different components within the Online Course Registration System. It provides a blueprint for system development and helps developers understand how different modules communicate and operate together.

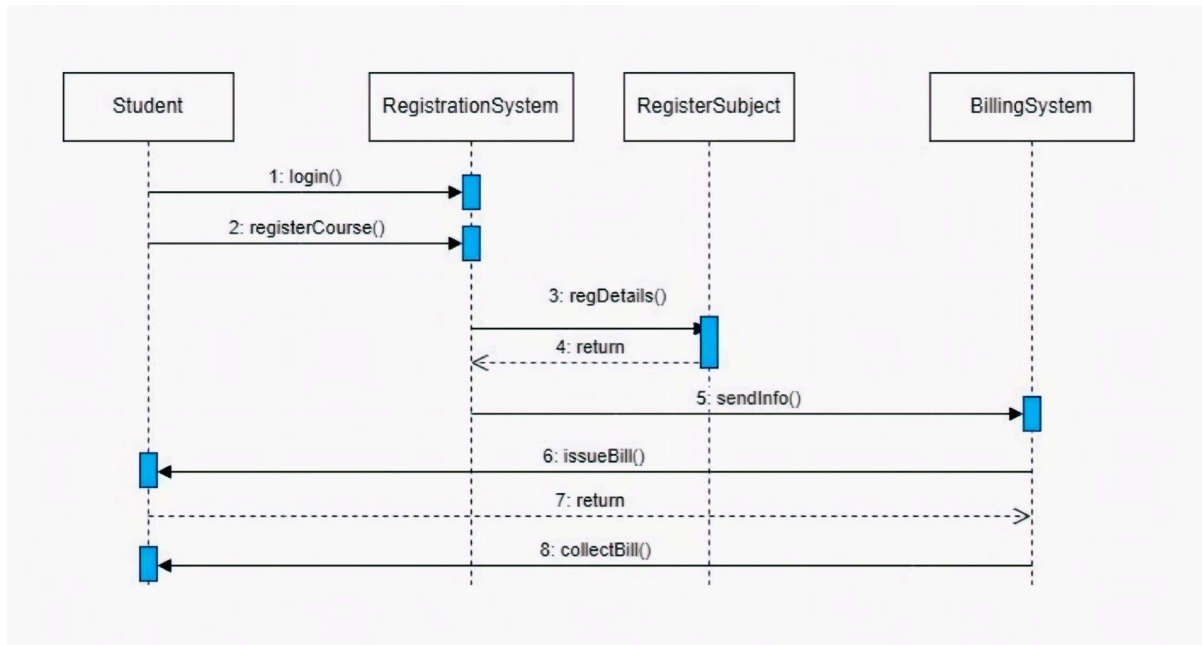
4.1 Class Diagram

The Class Diagram represents the static structure of the system by identifying classes, attributes, methods, and relationships among objects. It helps in understanding the object-oriented design of the Online Course Registration System.



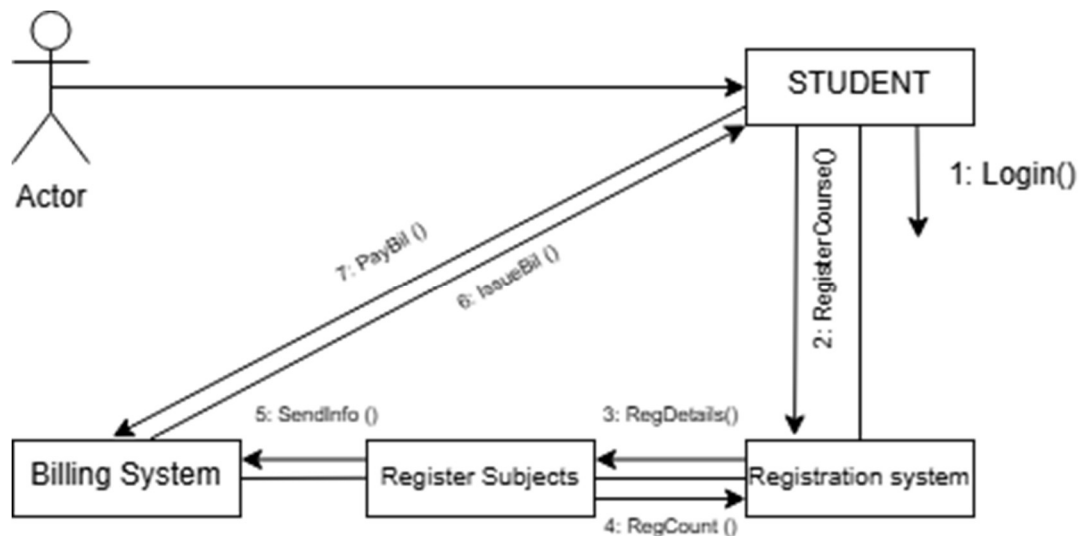
4.2 Sequence Diagram

The Sequence Diagram illustrates the sequence of interactions between users and system components over time. It shows how messages are exchanged during processes such as course registration and payment handling.



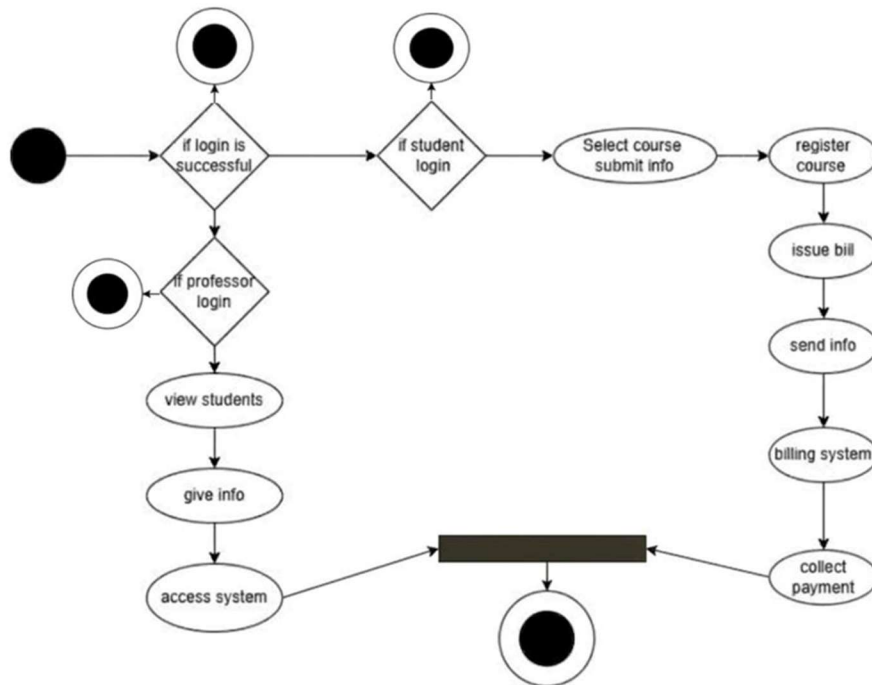
4.3 Collaboration Diagram

The Collaboration Diagram represents communication and coordination between system objects. It focuses on how different objects interact with each other to complete specific tasks within the system.



4.4 Activity Diagram

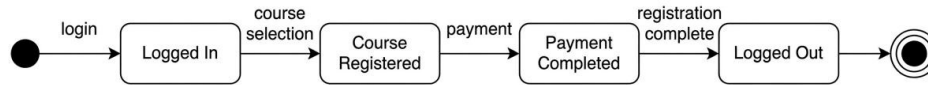
The Activity Diagram describes the workflow and operational processes of the system. It visually explains the step-by-step activities involved in tasks such as student registration and course management.



4.5 State chart Diagram

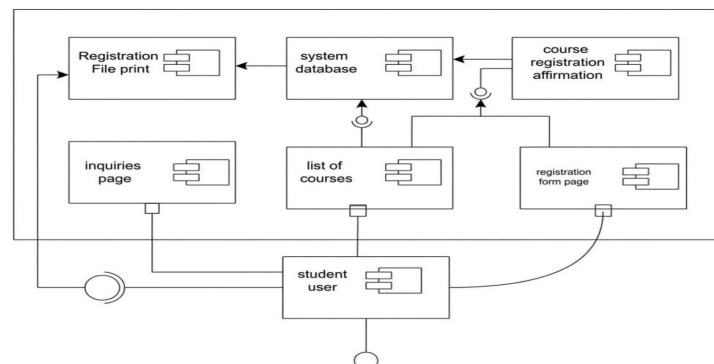
The State Chart Diagram represents the different states of an object or system and the transitions between those states during its lifecycle. It helps in understanding how the system behaves in

response to various events and actions.



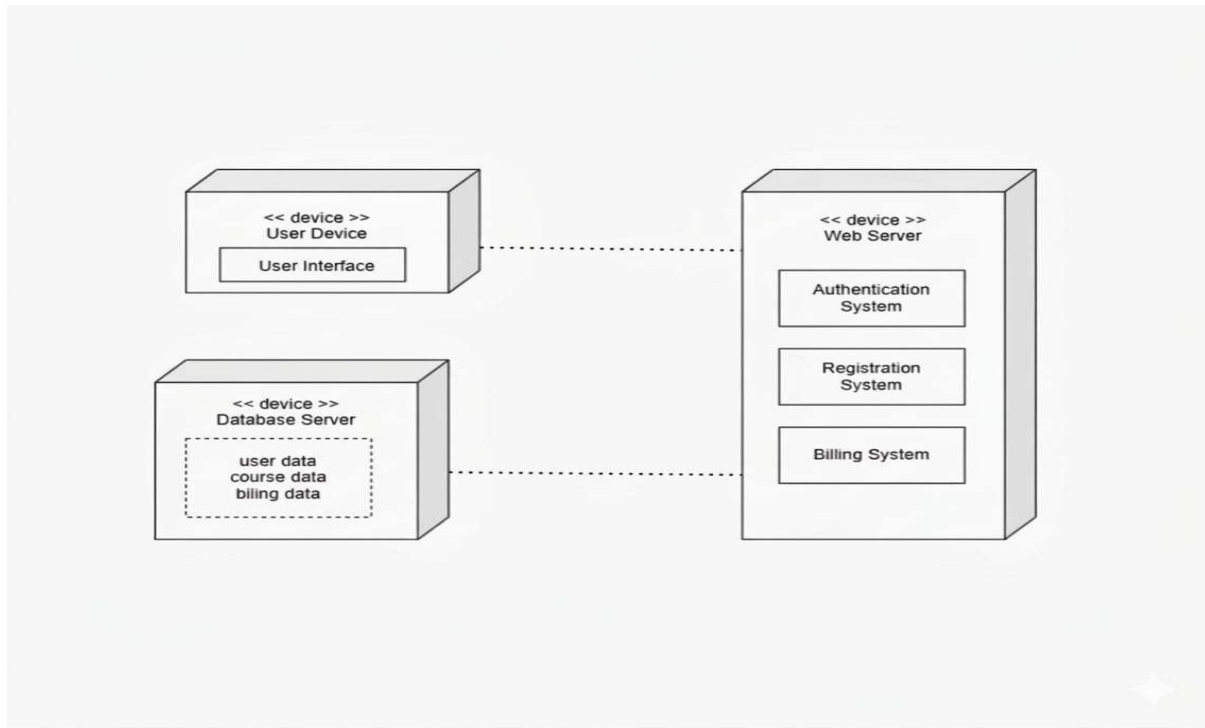
4.6 Component Diagram

The Component Diagram represents the high-level structure of the Online Course Registration System by showing its main software components and the relationships between them. It focuses on how the system is divided into different modules such as user interface, registration, billing, and database management. This diagram helps in understanding how different parts of the system are organized and interact with each other.



4.7 Deployment Diagram

The Deployment Diagram describes the physical architecture of the Online Course Registration System. It shows how the software components are deployed on hardware nodes such as client devices, web servers, application servers, and database servers. This diagram helps in understanding how the system will run in a real-world environment and how different devices and servers communicate with each other.



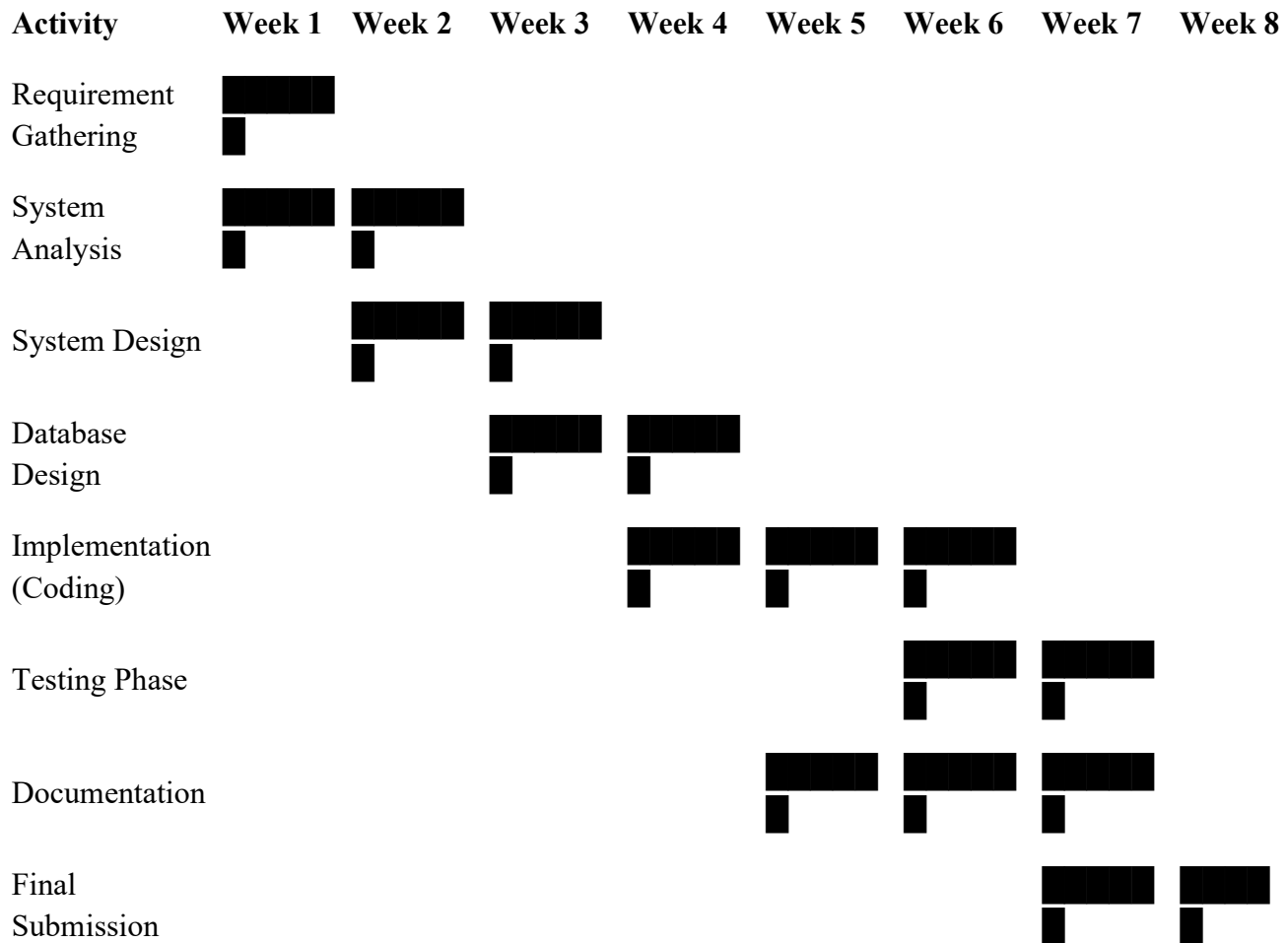
5. Overall Project Management

The Overall Project Management section describes the planning, scheduling, and execution of the Online Course Registration System project. It ensures that all development activities are properly organized, monitored, and completed within the given time frame.

5.1 Gantt Chart

The Gantt Chart represents the complete project timeline and shows how different phases of the Online Course Registration System are distributed over the semester. It helps in tracking progress, managing tasks, and ensuring timely delivery of the project.

Project Timeline (Estimated 8 Weeks)



Explanation

- **Weeks 1–2:** (Muhammad Afaan) Requirements are gathered and analyzed from users and stakeholders.
- **Weeks 2–3:** (Muhammad Afaan) System design and architecture are finalized.
- **Weeks 3–4:** (Sheraz Ahmad) Database structure is designed.
- **Weeks 4–6:** (Eman Shakeel) System implementation and coding are carried out.
- **Weeks 6–7:** (Ruhaab Maqsood) Testing and bug fixing are performed.
- **Weeks 5–7:** (Muhammad Afaan) Documentation is prepared in parallel with development.

- **Weeks 7–8:** (Muhammad Afaan, Sheraz Ahmad, Eman Shakeel, Ruhaab Maqsood) Final polishing and project submission are completed.

This Gantt Chart ensures smooth workflow, proper task distribution among team members, and timely completion of the Online Course Registration System project.